How We Built It: Domain-Adaptive Skin Cancer Classification, Explained

Oct 17, 2026 · @Deepak · Results: full data

This document explains the whole project, including what we asked, what data we used, how the model works, what every experiment found and what comes next. It's written so that someone with no machine-learning background can follow it and then explain it to others. It's a companion to the semester plan of 10 Oct 2026; where the two differ, this document reflects what was actually built.

Member	HT No	Branch
Ankita Vemavarapu	SE23UECM007	ECM
Aeleni Deepak Reddy	SE23UARI005	AI
Ranganaini Sai Ruchir	SE23UECM043	ECM

Supervisor: Dr. Anjli Varghese · Co-supervisor: Venkatasubramanian K. This is a research prototype, never a diagnostic tool.

The one-minute version

AI skin cancer tools are mostly trained on photos of lighter skin. We wanted to know how much worse such a model does on darker skin and on Indian patients, and whether common fixes help.

We trained a model on lighter-skin photos and tested it on several outside datasets. On the darkest skin types, photographed at a hospital it had never seen, it did **barely better than a coin flip** and missed over a third of cancers, four times as many as on light skin at the same hospital.

We tried five standard fixes, then the best of them on validation. **None closed the gap** at an outside hospital. The one thing that clearly helped was starting from a model **pre-trained on a large, varied set of dermatology photos** (Google's Derm Foundation). Even that still did worse on dark skin.

On everyday Indian clinic photos (fungal infections, vitiligo, acne), the models wrongly call many of them cancer. A real tool would first need a step that says "this isn't a tumour at all".

The problem, and why it matters

Skin cancer and AI

There are three main skin cancers:

- **BCC** (basal cell carcinoma): the most common, and rarely deadly.
- **SCC** (squamous cell carcinoma): more dangerous, and it can spread.
- **Melanoma**: the deadliest.

Each has harmless **lookalikes**, such as moles, seborrhoeic keratoses ("barnacles of ageing") and dermatofibromas. Telling a cancer from its lookalike is the hard part.

Many papers claim AI can do this about as well as dermatologists. But almost all of the training photos come from the US, Europe and Australia, which means mostly lighter skin. A review of public skin cancer datasets found skin type was recorded for only about 2% of images (Wen et al., 2022).

Skin type: the Fitzpatrick scale

Dermatologists describe skin with the **Fitzpatrick Skin Type (FST)** scale, from I to VI:

- **I–II:** very fair, always burns.
- **III–IV:** medium to olive or light brown. Most Indians are roughly **IV–V**.
- **V–VI:** brown to deeply pigmented.

In this project, **"source" means FST I–III**, the skin we train on. **"Target" means FST IV–VI**, the skin we want the model to work well on.

Why the gap is a real problem

On darker skin, lesions look different. Redness is harder to see, pigment contrast is different, and cancers are often found later. A model that only learned "cancer on pale skin" may miss cancers on brown skin, and a missed cancer is the dangerous kind of error. This is the **domain shift** problem: a model learns in one setting (the "domain") and is used in another.

> **Analogy:** you learn to identify birds from a European field guide, then go birdwatching in India. Some birds are the same, but the light, the backgrounds and the species mix all differ. You'll make more mistakes, even though you "know birds".

India adds a second mismatch. Indian hospital series are dominated by BCC and SCC, while melanoma, the focus of most public data, is a minority and mostly *acral* (palms, soles, nails).

Research questions

#	Question	Experiment

RQ1	How big is the gap between light and dark skin, and which cancers get mistaken for which lookalikes?	E1, E2
RQ2	Do epidemiology-guided choices help? That means cleaning up the image for darker skin, and telling the model which cancers are common in India.	E3, E4
RQ3	Does adding hand-measured texture information help?	E6
RQ4	How much dark-skin labelled data closes the gap? How much does "statistical alignment" close with no labels at all?	E5, E7
—	Does the best fix, chosen on validation, do better on a second network?	E8

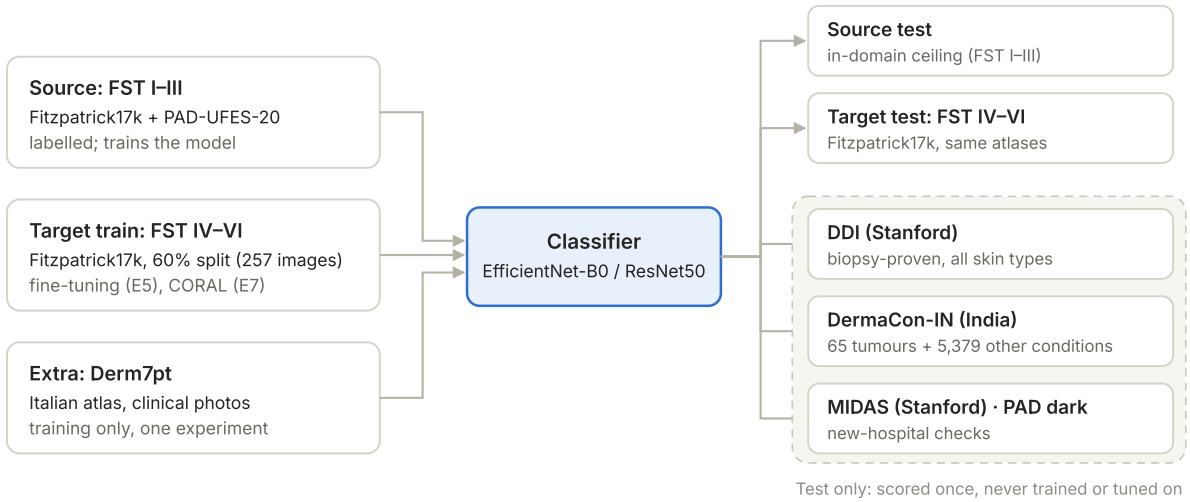
Vocabulary, in plain English

Term	What it means
Model / classifier	A program that looks at a photo and outputs four probabilities, e.g. *BCC 10%, SCC 70%, melanoma 5%, benign 15%*. The biggest one is its answer.
Training	Showing the model thousands of labelled photos so it adjusts its internal numbers to get more of them right.
Train / validation / test	The *textbook* it studies, the *practice exams* used to make decisions (when to stop, which version is best), and the *final exam* used only to report results. Peeking at the final exam while studying makes the score meaningless. Most of our rules exist to prevent that.
AUC	Our main score. Pick one random cancer photo and one random non-cancer photo: AUC is the chance the model rates the cancer as more suspicious. **1.0 is perfect and 0.5 is a coin flip.** Roughly, 0.9 is excellent, 0.8 good, 0.7 fair and 0.6 weak.
Binary vs macro AUC	Binary means cancer vs not-cancer. Macro is the average of "BCC vs rest", "SCC vs rest", "melanoma vs rest" and "benign vs rest".
Missed-cancer rate	The share of real cancers the model calls benign. This is the dangerous error.
False-alarm rate	The share of harmless lesions the model calls cancer. It causes anxiety, unnecessary biopsies and overloaded clinics.
Confidence interval (CI)	`0.50 [0.40, 0.60]` means our best estimate is 0.50, and the true value probably lies between 0.40 and 0.60. Small test sets give wide intervals. **If two methods' intervals overlap heavily, we can't claim one is better.**
Seed	Training involves randomness. We train every model three times (seeds 0, 1 and 2) and average, so one lucky run can't fool us.

Pre-trained / transfer learning	Instead of a blank model, we start from one that already learned to see from about a million everyday photos (ImageNet: dogs, cars, cups). It already knows edges, textures and shapes; we then teach it skin lesions.
Fine-tuning	Taking an already-trained model and training it a bit more on new data.

The data

Models learn on light skin; darker-skin and Indian images adapt and test them



data roles · 3 training inputs, 6 test sets

The datasets, and why each one is there

Dataset	Role	Why it's there
Fitzpatrick17k	Training (FST I–III) and target train/val/test (FST IV–VI)	About 16,600 photos from online dermatology atlases, each labelled with a diagnosis and a skin type. We use only the cancer and lookalike images.
PAD-UFES-20 (Brazil)	Training (FST I–III). Its 60 FST IV–VI images form a test-only set.	Biopsy-confirmed smartphone photos from a Brazilian clinic.
DDI (Stanford)	Test only. **The most important test.**	656 images, every lesion biopsy-proven, all skin types. It comes from a *different hospital*, with real clinic photos rather than textbook ones. Its licence forbids showing its images, so we only report numbers.

DermaCon-IN (India)	Test only	Only 65 are cancers or lookalikes (19 cancers, 46 lookalikes). The other 5,379 are everyday conditions: tinea, vitiligo, scabies, acne, psoriasis, eczema. We use them to ask how often the model wrongly cries "cancer" on ordinary Indian patients.
MRA-MIDAS (Stanford)	Test only	1,380 biopsy-proven photos, 96% lighter skin. It measures the effect of changing hospital *without* changing skin tone.
Derm7pt (Italy)	Extra training data, one experiment	About 950 clinical photos, added because our training set had very few melanomas and benign lesions.

The full image set. Many of Fitzpatrick17k's images originally came from a website whose links are now dead. We asked the authors, and they sent the complete set of 16,577 images. Every result in this document uses it. Earlier runs on a partial download (1,919 of 4,491 relevant images) were development runs only: they helped us find and fix protocol problems, but none of their numbers are reported.

Why most sets are test-only. We train only on Fitzpatrick17k and PAD-UFES-20 (plus Derm7pt in one experiment). DDI, DermaCon-IN, MIDAS and PAD's dark-skin images are *never* trained on, because they represent the real question: does the model work somewhere it has never been?

Cleaning the data: a five-step pipeline

Raw medical datasets are messy. Before any training, five steps run once, and their outputs are frozen in the repository. Nobody re-splits afterwards, so every experiment uses exactly the same data.



1. Label map

The datasets use 460 different diagnosis names, such as "superficial spreading melanoma", "morpheiform BCC", "dermatosis papulosa nigra" and "Bowen's disease". Each one has to go into one of our classes:

- **bcc / scc / mel:** every subtype of each cancer. Bowen's disease is SCC at its earliest stage, so it counts as SCC.
- **benign:** true lookalikes only, i.e. moles, seborrhoeic keratosis, dermatofibroma, solar lentigo and pyogenic granuloma.
- **exclude:** rare cancers (e.g. Kaposi sarcoma), non-tumours, and things that aren't really moles despite the name.

Code rules make a first pass, then humans review every row. The first pass is done; the team still needs to confirm it.

A special case: actinic keratosis (AK). AK is a *pre-cancer*, so calling it either "benign" or "cancer" would be medically wrong. We never train on it. Instead, we test the models on 311 AK images as a side analysis. The baseline calls 90% of them malignant, which is understandable, since AK really can turn into SCC.

2. Manifest

One big table with one row per image: label, skin type, patient ID, body site, source dataset, and the exclusion reason if there is one. Images with **unknown skin type are dropped**, because we can't study skin type without knowing it.

3. De-duplication

The same mole is often photographed twice, re-cropped, or watermarked and reposted. If one copy lands in training and the other in the test, the model simply remembers it, which is peeking at the final exam. Images are grouped when any of these holds:

- same patient;
- near-identical pixels (a perceptual-hash fingerprint differs by at most 2);
- at least 90% similarity in the model's internal description of the two images.

The thresholds were chosen by inspecting real pairs by eye. At looser settings, unrelated "small brown spots" started matching.

4. Splits

Light skin is split 70 / 15 / 15 (train / validation / test). Dark skin is split 60 / 20 / 20. A duplicate group is never split across two sets, and the code checks this automatically.

Set	Images	BCC	SCC	Melanoma	Benign
Source train (light)	2,022	902	400	337	383
Source val / test (light), each	435	194	86	72–73	82–83
Target train (dark)	257	65	70	61	61
Target val / test (dark), each	86	22	23–24	20–21	20

Notice how **small** the dark-skin sets are: 86 test images, 20 of them benign. That's why the intervals on those numbers are wide.

5. Count check

If the dark-skin test set held fewer than 15 SCC images, we would merge BCC and SCC into one class, because scores on so few examples are just noise. There are 23, so we kept four classes.

The model, and how it's trained

What the model is

We use **EfficientNet-B0**, a well-known, compact image-recognition network (a convolutional neural network, or CNN), pre-trained on ImageNet. We replace its original 1,000-way "dog / car / cup" output with a small final layer that outputs our four probabilities. To check the results aren't a quirk of one architecture, we repeat the main experiment with a second, older network, **ResNet50** (E2).

> **How a CNN sees:** early layers detect edges and colour blobs, middle layers detect textures and shapes, and late layers detect whole concepts. The final layer turns all of that into "how much does this look like BCC, SCC, melanoma or benign?"

Training in two stages

Stage	What is trained	Settings
1	Only the new final layer; everything else is frozen. It learns the four-class task quickly without disturbing what the network already knows.	5 epochs (full passes over the data), learning rate 1e-3
2	The top 3 blocks of the network are unfrozen and gently adapted to skin.	Up to 30 epochs; head 1e-4, backbone 3e-5; stops early if no improvement for 5 epochs

The learning rate is how big each adjustment is. We tested three values for stage 2 on validation data and chose 3e-5. The plan's 1e-5 hadn't finished learning within the time budget.

Tricks used during training

- **Data augmentation.** Each time an image is shown, it's randomly altered: cropped to 70–100% of its area, flipped either way, rotated up to 30°, with brightness and contrast nudged by ±10%. There's no hue change, because that would alter apparent skin tone. This stops the model memorising exact photos.
- **Class balancing.** We have 902 BCC images but only 337 melanomas. Left alone, the model would learn "when in doubt, say BCC". Mistakes on rare classes are therefore weighted more heavily, as if every class were equally common.
- **Early stopping on the light-skin validation set.** We stop when the practice score stops improving and keep the best version. The baseline must never have seen a dark-skin label. The plan's original choice, the dark-skin validation set, would have quietly leaked that information in.
- **Three seeds** for everything, averaged.
- **Hardware:** it all runs on an ordinary laptop CPU with no graphics card, at about 44 images per second.

How we keep the results honest

These rules are a big part of the project's value. Many published results are inflated by quietly breaking them.

1. **Decisions are made on validation data, never on test data.** That covers when to stop, which settings to use, and what goes into the combination. Every such choice is recorded with its reason in `DECISIONS.md` (decisions D1–D23).
2. **The external tests can be taken only once.** DDI, DermaCon-IN, MIDAS and PAD-dark are scored by a special command that writes each score to a ledger and *refuses to score the same model on the same test twice*. Nobody can try, look, tweak and retry.
3. **Intervals respect patients.** Confidence intervals come from a bootstrap: the test set is re-drawn 1,000 times to see how much the score wobbles. It re-draws *whole patients or duplicate groups*, so five photos of one person don't count as five independent pieces of evidence.
4. **A fixed operating point.** Someone has to choose the probability cut-off for "call it cancer". We set it so the model catches 90% of cancers on light-skin validation photos, then apply *that same cut-off everywhere*. This mirrors real deployment, and it shows whether a cut-off still works at a new hospital.
5. **Pilot runs disclosed.** Early runs that checked the code worked did look at some test scores. We recorded this and made sure no decision used them.

> **Analogy for the operating point:** you set a smoke alarm's sensitivity in your own kitchen. Then you install it in a different house. Does it still go off when it should?

The experiments, one by one

The key number throughout is **DDI V–VI**: the darkest skin, at a hospital the model has never seen. All numbers are means over three seeds, with 95% intervals in brackets where they matter.

E1 + E2 · The baseline: how big is the gap? reference

Train on light skin only, then test everywhere.

Test set	AUC	Cancers missed (fixed cut-off)
Light skin, same sources (source test)	0.89 (macro)	10%
Dark skin, same atlases (Fitzpatrick17k IV–VI)	0.89 (macro)	14%
DDI FST I–II (new hospital, light)	0.65	9%
DDI FST III–IV	0.66	16%
DDI FST V–VI (new hospital, dark)	**0.53 [0.43, 0.62]**	**36%**
MIDAS (new hospital, light skin)	0.69	40%
DermaCon-IN (India)	0.71	26%

- At the fixed cut-off, the baseline misses **36% of cancers on DDI V–VI**, four times the 9% it misses on DDI I–II. At DDI it also raises false alarms on 66–71% of benign lesions, against 31% at home.
- ResNet50 (E2) is weaker at home (0.79 vs 0.89) but a little better at DDI. It still drops from FST I–II (0.75) to V–VI (0.58), so the gap isn't specific to one architecture.
- **There is no gap inside Fitzpatrick17k** (0.89 vs 0.89), because its dark-skin photos come from the same textbook atlases as the training photos.
- **The gap appears at a real outside hospital.** That's why DDI matters more than the in-dataset split.
- This matches Daneshjou et al. (2022), who saw 0.72 → 0.57 on DDI for a different model.

E3 · Image processing: can we "clean up" the photos? no gain

Maybe the model is thrown by lighting and contrast rather than by skin itself. We tested three classic fixes, each on its own, and then all together:

- **Colour constancy (Shades-of-Gray):** removes colour casts from lighting, so a yellow-tinted indoor photo looks as if it were taken in neutral light.
- **CLAHE:** boosts local contrast on brightness only, so skin colour isn't distorted. The hope was that lesion edges on dark skin would stand out.
- **Lesion crop:** automatically zooms in on the spot. We tuned it so rulers, dark backgrounds and hair no longer confuse it.

Result:

- None beat the baseline on validation. Colour constancy tied (0.917 vs 0.916); CLAHE, the crop and all three together scored about 0.90.
- On DDI V–VI, every variant stayed between 0.51 and 0.55, inside the noise.

- The crop actually hurt on clinic photos (India 0.71 → 0.62, Brazil dark skin 0.83 → 0.77), probably because it cuts away the skin around the lesion.

Subtle extra finding: CLAHE missed fewer dark-skin cancers at DDI at the fixed cut-off (26% vs 36%), but its AUC barely moved (0.55 vs 0.53). It shifts the model's scores rather than making it better at telling cancer from benign.

What it means: the problem can't be fixed by adjusting colour or contrast. The difficulty with dark skin is deeper than lighting.

E4 · Indian cancer statistics: tell the model what's common here no gain

Published Indian hospital series show a cancer mix of roughly **BCC 45 : SCC 40 : melanoma 15**, which differs from our training data. If the model "expects" the Indian mix, maybe it does better on Indian patients. We tried this two ways:

1. Retrain with the loss weighted to that mix.
2. Mathematically adjust an already-trained model's outputs (a "prior shift").

We also tried the mixes from three regional series: Malwa-Punjab, Central India and Manipur.

Result: the validation score didn't move (0.915 retrained, 0.92 with the prior shift, vs 0.916). India (0.71 → 0.73–0.74) and MIDAS (0.69 → 0.70–0.71) went up slightly, inside the noise, and DDI V–VI not at all (0.51–0.52). The two approaches agree, and all four regional mixes agree. Fewer Indian non-cancer photos were called cancer at the default decision (47% → 41%), but the same 43% at the fixed cut-off.

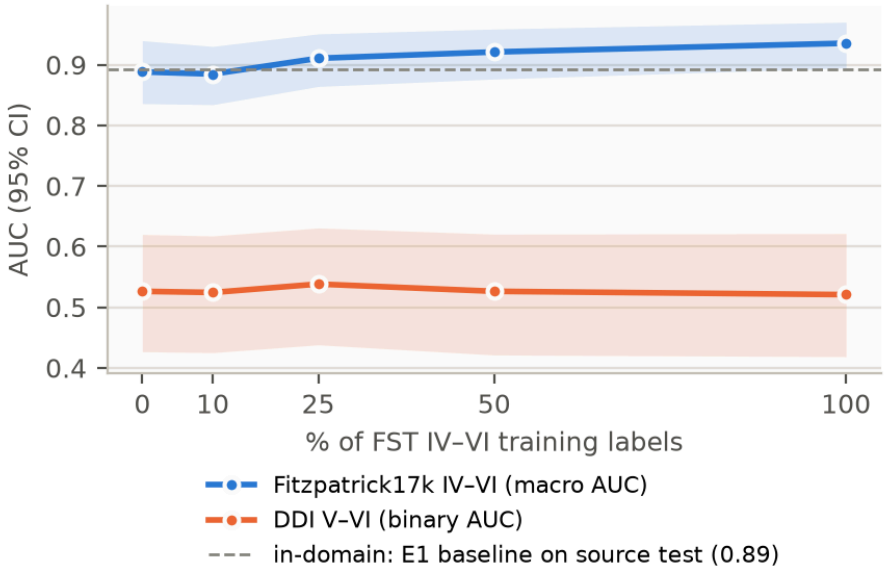
What it means: the expected mix shifts the model's guesses *between* cancer types (more SCC, less melanoma). It doesn't help separate *cancer from benign* on dark skin, which is the real problem.

E5 · Show it some dark skin: how much data does fine-tuning need? helps in-domain only

We took the baseline and trained it a bit more on 10%, 25%, 50% and 100% of the 257 dark-skin training images, to draw a learning curve.

A bug we found and fixed. In the development run, the 10% subset (17 images) was smaller than one batch of 32. The code dropped incomplete batches, so the model got **zero training steps**, and the "10% result" was really just the baseline. We spotted it in the training logs, not from test results. We gave every fraction the same number of training steps and re-ran it.

Dark-skin images used	0	28	67	130	257
Same-atlas dark-skin test	0.89	0.89	0.91	0.92	**0.94**
DDI V–VI (outside hospital)	0.53	0.52	0.54	0.53	0.52



E5 · the same-atlas score climbs with more dark-skin labels; the outside-hospital score stays at chance

What it means: this is one of the most important results.

- Dark-skin images **help on photos like them**, from the same atlases in the same style.
- They **don't help at a different hospital**. The model learned what these atlases' dark-skin photos look like, not what cancer looks like on dark skin.
- At the fixed cut-off, false alarms on Indian non-cancer photos rose from 43% to 51%, and the light-skin validation score dropped by 0.02.

> **Analogy:** studying past papers from one teacher makes you great at that teacher's exams. It doesn't make you good at the subject.

E6 · Hand-made texture features: does the network miss texture? no gain

Dermatologists look at texture: scaliness, network patterns, roughness. We computed **78 classic texture numbers** per image:

- **LBP:** tiny repeated patterns, from how each pixel compares with its neighbours.
- **GLCM:** how often bright and dark pixels sit next to each other, i.e. smoothness vs roughness.
- **Gabor filters:** stripes and ridges at different angles and sizes.
- **Colour statistics:** the average, spread and skew of each colour channel.

We joined these to the network's own 1,280-number description of the image and trained a small classifier on top. We also trained a **fair control** that used only the network's description. Without it, any gain could just come from the new classifier rather than from texture.

Result:

- Texture added almost nothing: fusion scored 0.914 on validation against 0.906 for the control and 0.916 for the baseline.

- Texture alone was weak (0.73 on validation). But it lost less at outside sites (Brazil dark skin 0.85, DDI V–VI 0.55), a hint that texture depends less on skin tone and camera. The interval includes chance, so this is only a hint.

E7 · Deep CORAL: align light and dark images without labels

small outside-site trend

Inside the network, each image becomes a list of 1,280 numbers. If light-skin and dark-skin images produce differently shaped "clouds" of these numbers, the decision layer tuned on light skin will misjudge dark skin.

CORAL adds a penalty during training that pushes the two clouds towards the same spread and correlation pattern (the same covariance). Crucially, it needs dark-skin *images* but not their diagnoses, and unlabelled photos are much easier to get than biopsy-confirmed ones.

A fix to the plan. The plan's penalty strengths (λ = 0.1, 1, 10) would have made the penalty about 0.1% of the training signal, so it would effectively have done nothing. We used the CORAL paper's own balancing rule instead, which gives $\lambda \approx 1000$, and tested 100, 1000 and 10000.

Result:

- No setting beat the baseline on validation (best 0.914 at λ = 10000, vs 0.916).
- On outside sets there's a small, consistent trend. Stronger alignment raised Brazil dark skin from 0.83 to 0.88, MIDAS from 0.69 to 0.72 and DDI V–VI from 0.53 to 0.55. It is the only planned method that moved all three outside sets the same way, but every interval still overlaps the baseline's.
- We also checked whether CORAL made light and dark images harder to tell apart. A simple test (a domain probe) could tell them apart **67%** of the time before CORAL (50% would be chance) and **70%** after. CORAL did not hide the domain difference. Fine-tuning on dark-skin labels (E5) narrowed it a little, to 62%.

E8 · The best choice on validation

worse outside

The plan was to combine every component that beat the baseline *on validation*. On the full data only E5 fine-tuning did, so E8 is fine-tuning alone. On EfficientNet that is exactly the E5 run above; we also ran it on ResNet50.

Result: on ResNet50, fine-tuning raised the same-atlas dark-skin score (0.79 → 0.85) and *lowered* every outside clinic: DDI light skin 0.75 → 0.67, DDI dark skin 0.58 → 0.56, India 0.76 → 0.70, MIDAS 0.65 → 0.61. **The best choice on validation does not close the gap at another hospital, and can widen it.** Choosing on photos from the same atlases rewards learning those atlases.

Extra experiments: stronger baselines one lever works

Once the planned methods failed, we asked what does move the needle.

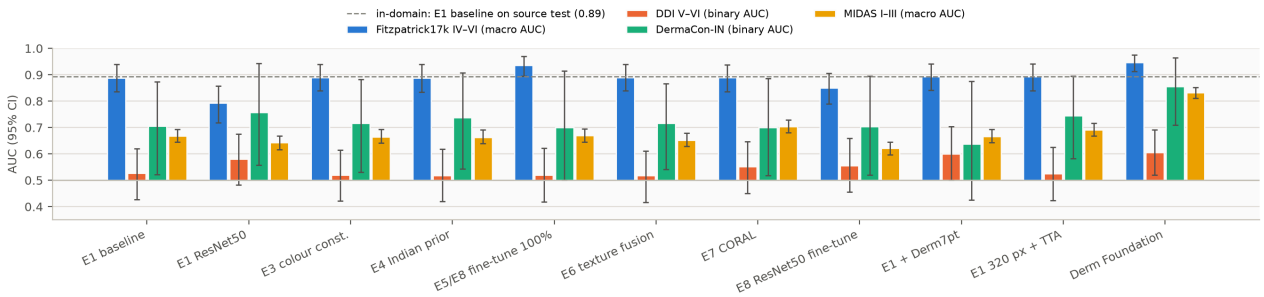
Extra	What it is	DDI V–VI	Takeaway
Seed ensemble	Average the three seeds' predictions	0.53	The seeds already agree (changes of 0.004 at most); randomness isn't the problem
320 px + flip TTA	Bigger images; at test time, average over 4 flipped versions	0.53	Helps lighter-skin outside sites a little (India 0.71 → 0.74, MIDAS 0.69 → 0.71), not dark skin
+ Derm7pt	About 950 extra Italian training photos	0.60	Best bump among CNNs, but more false alarms on Indian photos (43% → 60%): its benign photos are almost all moles
Derm Foundation	Google's dermatology model, kept frozen; we train only one simple layer on top	**0.61 [0.52, 0.69]**	**Best on every outside set**

Test set	Our baseline	Derm Foundation
DDI FST I–II	0.65	**0.78**
DDI FST III–IV	0.66	**0.83**
DDI FST V–VI	0.53	**0.61** (interval above chance)
DermaCon-IN (India)	0.71	**0.86**
MIDAS (light skin, new hospital)	0.69	**0.82**

Why it works: it was pre-trained on a large, varied set of skin photos rather than on dogs and cars. It has seen far more cameras, clinics and skin tones than our 2,022 training images ever could.

Caveats:

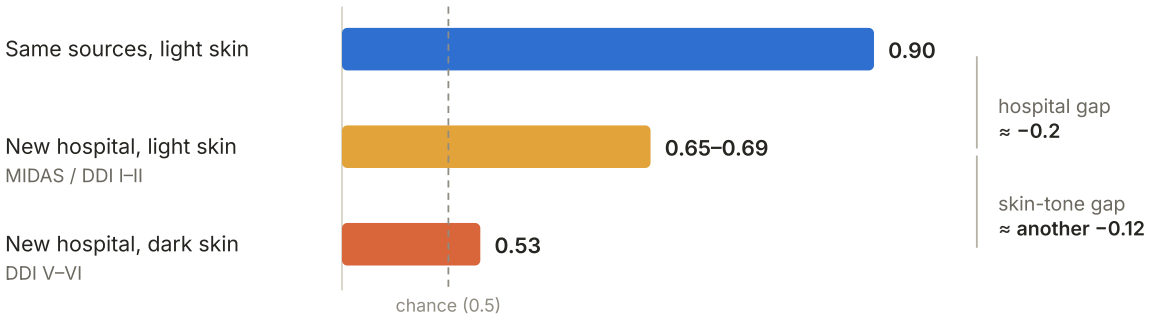
- It may have seen the *public* datasets during its own training, so those scores could be flattering. DDI and MIDAS are access-controlled and DermaCon-IN came out later, so those three are trustworthy.
- The skin-tone gradient remains (0.78 → 0.83 → 0.61).
- Its cut-off doesn't travel. At the fixed operating point it misses only 15% of DDI V–VI cancers, but calls 73% of Indian non-cancer photos malignant.
- Adding the atlas dark-skin images to it repeats E5's lesson: validation and India go up (0.86 → 0.90), but dark skin at DDI goes down (0.61 → 0.55).



Every method on four test sets, with 95% intervals. The dashed line is the in-domain baseline; 0.5 is chance. MIDAS is shown here as macro AUC (the text quotes binary AUC).

The big picture

1 · Two separate gaps, stacked on each other



binary AUC of the baseline · bars start at 0.4

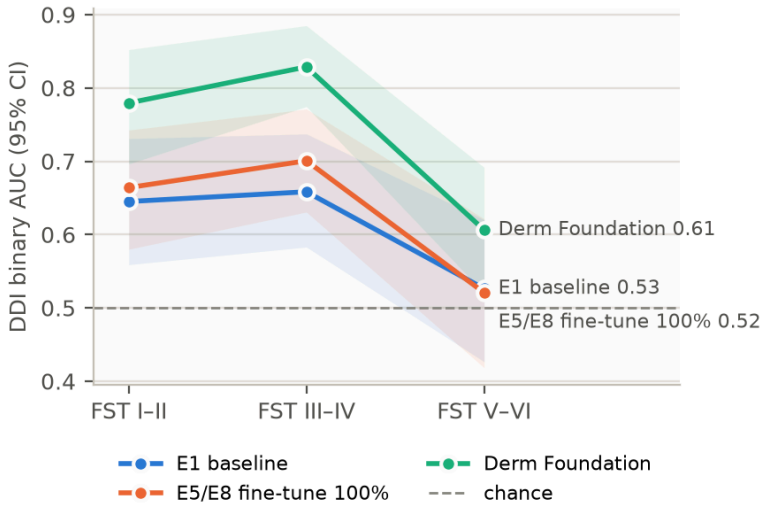
MIDAS makes this visible. It's a new hospital with mostly light skin, so it isolates the effect of the hospital itself: camera, lighting, photographer and patient mix. Most "dark skin" fixes only target the second gap, and they did so with dark-skin data from the *same* atlases. So they couldn't touch the first.

2 · What doesn't work, and why that's useful

Image clean-up, Indian statistics, texture features, CORAL alignment and the best-on-validation choice (fine-tuning) all failed at outside hospitals. That's a meaningful **negative result**. These are exactly the methods many papers propose, and under strict no-peeking rules they don't solve the real-world version of the problem.

3 · What does work

Better pre-training (Derm Foundation) is the one lever that moved every outside set. Diversity of experience beats clever tricks.



DDI by skin type · Derm Foundation lifts every bucket, but the drop to FST V–VI remains

4 · The "only four boxes" problem

The model knows only four answers: BCC, SCC, melanoma or benign tumour. Show it a fungal infection and it *must* pick one of the four, often a cancer. On 5,379 everyday Indian clinic photos, at the default decision the baseline calls **47%** malignant and Derm Foundation **78%** (43% and 73% at the fixed cut-off). Any real tool needs a first step that says "not a tumour / out of scope" before the cancer classifier runs.

> **Analogy:** ask someone who only knows four dog breeds to identify a cat. They'll confidently name a dog breed.

5 · A cut-off set at home doesn't travel

A cut-off that catches 90% of cancers at home catches only 64% on dark skin at DDI, and there it raises false alarms on 66–71% of benign lesions, against 31% at home. Scores shift between sites, so real use would need re-calibration at each clinic.

Honest limitations

Limitation	Why it matters
Small dark-skin test sets	86 target test images (20 benign); DDI V–VI has 207 images (48 cancers). Hence the wide intervals, and why we avoid claiming small differences.
Estimated skin type	Fitzpatrick types are judged from photos, not measured on patients.
Thin training classes	337 melanomas and 383 benign images in training, against 902 BCC.
Possible pre-training overlap	Derm Foundation may have seen the public datasets; only its DDI, MIDAS and DermaCon-IN scores are clean.

Acral lesions	About 50 in PAD-UFES-20, 39 in MIDAS and 9 in DDI (none flagged in Fitzpatrick17k), too few for firm conclusions.
Atlas labels	Fitzpatrick17k diagnoses are atlas-scraped, not biopsy-verified. DDI (biopsy-proven) is the clean check.

The future plan

To finish this semester

#	Task	Status
1	**Team confirms the label map** (460 diagnosis → class rows)	Second full review done (`docs/LABEL_MAP_REVIEW.md`): one 1-image error, no effect on scores; team sign-off open
2	**Check 100 lesion crops by eye** and record the failure rate	Done: about 15–20% of 100 crops miss the lesion
3	**Grad-CAM grid**: heat-maps showing *where* the model looks	Done (kept local: patient photos); findings in `docs/RESULTS.md`
4	**Mid-semester review pack**: data, splits, E1–E3 with intervals, gap figure, Grad-CAM	Done: `docs/MIDSEM_PACK.md`
5	Acral check	Done: counts and MIDAS scores; too few (39) for conclusions
6	**IEEE-format report, 15-minute presentation**, one-command reproduction on a clean machine	Report and talk open; clean-machine check done
7	Update the semester plan with the protocol changes listed in `docs/PLAN_CHANGES.md`	Done: revised plan in `docs/SEMESTER_PLAN.md`

TIME-SENSITIVE

DDI access expires 12 Nov 2026. Any further DDI scoring must happen before then, or access must be renewed.

Research directions the results point to

1. **Make a dermatology foundation model the main backbone**, since it's the only lever that worked.
2. **Teach the "is this a tumour?" gate about rash-like cancers.** Our gate (E10) cut false alarms on everyday Indian conditions from 73% to 7%, but it also turned away 42% of real dark-skin

cancers at DDI (missed cancers 15% → 52%). Retraining it with dark skin (E10b) changed nothing, because most of those cancers are mycosis fungoides, a skin lymphoma that looks like a rash, not a lump. A usable gate must keep rash-like cancers, or send what it rejects to a doctor instead of clearing it.

3. **Calibrate per clinic:** use a small local sample to re-set the cut-off at each new site.
4. **Collect dark-skin data from many institutions**, not more from the same atlases. E5 shows that's what's missing.
5. **Follow up on texture** (E6's hint that it depends less on skin tone) with more DDI-like data.
6. **Beyond this project:** an Indian hospital collection with ethics approval as a true external test, a dedicated acral melanoma study, and DANN as a second adaptation method.

How to explain it to other people

The 30-second version

> "AI skin cancer tools are trained mostly on light skin. We trained one and found it drops to near coin-flip accuracy on the darkest skin at an unfamiliar hospital, missing over a third of cancers. We tested five popular fixes; none solved it at an outside hospital. What helped was starting from a model pre-trained on lots of varied dermatology photos. We also found these tools wrongly flag many ordinary Indian skin conditions as cancer, so they need an 'is this even a tumour?' step first."

The 2-minute version

1. **Problem:** AI dermatology data is mostly light-skinned, while India is mostly FST IV–V, and missed cancers are dangerous.
2. **Setup:** we train on light skin (Fitzpatrick17k + Brazil) and test on six sets, including a biopsy-proven Stanford set with all skin tones and Indian clinic photos. The rules are strict: choices on validation only, outside tests scored once.
3. **The gap:** 0.90 at home, about 0.65–0.69 at a new hospital, and 0.53 on dark skin at a new hospital. Two gaps, stacked.
4. **The fixes:** image clean-up, Indian cancer statistics, texture features, dark-skin fine-tuning, CORAL alignment, and the best of these on a second network. None closed the outside-hospital gap, and fine-tuning helped only on photos from the same source.
5. **What worked:** a dermatology foundation model, which improved every outside set (India 0.71 → 0.86, dark skin 0.53 → 0.61). A gap remains.
6. **Deployment warning:** false alarms on everyday conditions, and cut-offs that don't transfer between sites. A "not a tumour" gate removes most false alarms, but it also rejects rash-like cancers, which make up over half of DDI's darkest-skin cancers.
7. **Next:** a gate that keeps rash-like cancers, per-site calibration and multi-institution dark-skin data.

Questions people will ask

"So did the project fail, since the fixes didn't work?"

No. A carefully measured negative result is a finding. These fixes are widely proposed; we show, without test-set peeking, that they don't solve the real-world problem, and we identify what does help.

"Why does it score 0.89 on dark skin in one test and 0.53 in another?"

The 0.89 is on dark-skin photos from the same textbook atlases as training. The 0.53 is on real clinic photos from a different hospital. The atlas test mostly measures "same source", not "dark skin".

"Why not just add more dark-skin photos?"

We did (E5), and it helped only on photos from the same source. The data needs to come from many different clinics and cameras.

"Why are your intervals so wide?"

The dark-skin test sets are small (86 and 207 images). That's also why we don't over-claim small differences.

"Why not train on DDI or DermaCon-IN?"

Then there would be no honest outside test left. DDI's licence also restricts its use, and DermaCon-IN has only 65 relevant images.

"Is the Derm Foundation result cheating if it saw the data?"

Possibly for the public datasets, which we flag. DDI and MIDAS are access-controlled and DermaCon-IN post-dates the model, so those results stand.

"Is there a published model that does better?"

Not under the same rules. On DDI, the best models published without training on DDI score 0.72 on light skin and 0.57 on dark skin (Daneshjou et al., 2022). Our Derm Foundation model scores 0.78 and 0.61, level with a panel of three dermatologists on dark skin (0.62). Models that score higher were trained on part of DDI itself, which we deliberately never do.

"Why not combine the best parts of every model?"

We tried (E9). Network layers trained separately can't be spliced together, so we combined their predictions and their features. The best combination chosen on validation topped the home tests (0.97) but was worse than Derm Foundation alone at every outside clinic (dark skin 0.54 vs 0.61). Validation on the same atlases rewards learning those atlases.

"Why leave out actinic keratosis?"

It's a pre-cancer. Labelling it either "benign" or "cancer" would be medically wrong, so we score it separately.

"Could this be used in clinics now?"

No. It's a research prototype. It misses too many cancers on dark skin and raises too many false alarms on everyday conditions.

Where things live in the repository

You want...	Look at
The protocol rules and why they were chosen	`DECISIONS.md`
All results with commentary	`docs/RESULTS.md`
Setup and commands	`README.md`
Experiment settings, one file per experiment	`configs/` (all built on `base.yaml`)
Data cleaning	`src/skincancer/data/`
The network and CORAL loss	`src/skincancer/models.py`
Training	`src/skincancer/train.py`
Image clean-up (E3)	`src/skincancer/preprocess.py`
Texture features and fusion (E6)	`features/texture.py`, `fusion.py`
Derm Foundation	`foundation.py`, `scripts/derm_foundation_embed.py`
Scoring, intervals, prior shift	`metrics.py`, `evaluate.py`
One-time external scoring and its ledger	`predict.py`, `runs/final_ledger.csv`
Figures	`reports/figures/`
Run everything	`scripts/run_all.sh`, `scripts/run_extras.sh`

One design idea worth knowing: every model just writes a spreadsheet of predictions (an image ID plus four probabilities). All scoring, intervals and figures read only those spreadsheets. Adding a new method means producing that spreadsheet, and everything else works automatically.

Key references

- Daneshjou et al. (2022), *Disparities in dermatology AI performance on a diverse, curated clinical image set*, Science Advances. The DDI dataset.
- Groh et al. (2021), *Evaluating deep neural networks trained on clinical images in dermatology with the Fitzpatrick 17k dataset*, CVPR Workshops.

- Pacheco et al. (2020), *PAD-UFES-20*, Data in Brief.
- Madarkar et al. (2025), *DermaCon-IN*, NeurIPS Datasets & Benchmarks.
- Kawahara et al. (2019), *Seven-point checklist and skin lesion classification* (Derm7pt), IEEE JBHI.
- Wen et al. (2022), *Characteristics of publicly available skin cancer image datasets*, Lancet Digital Health.
- Sun & Saenko (2016), *Deep CORAL: Correlation Alignment for Deep Domain Adaptation*, ECCV Workshops.
- Supekar et al. (2021), *Clinical spectrum of cutaneous malignancies in central India*, Indian Journal of Dermatology; plus the Malwa-Punjab and Manipur hospital series.
- Google Health, *Derm Foundation* model card (Hugging Face: google/derm-foundation).